

DS2278 Features



IMPORTANT

DO NOT use the CR2278-PC cradle with any scanners other than the DS2278 digital scanner. No other cradles are compatible with the DS2278.

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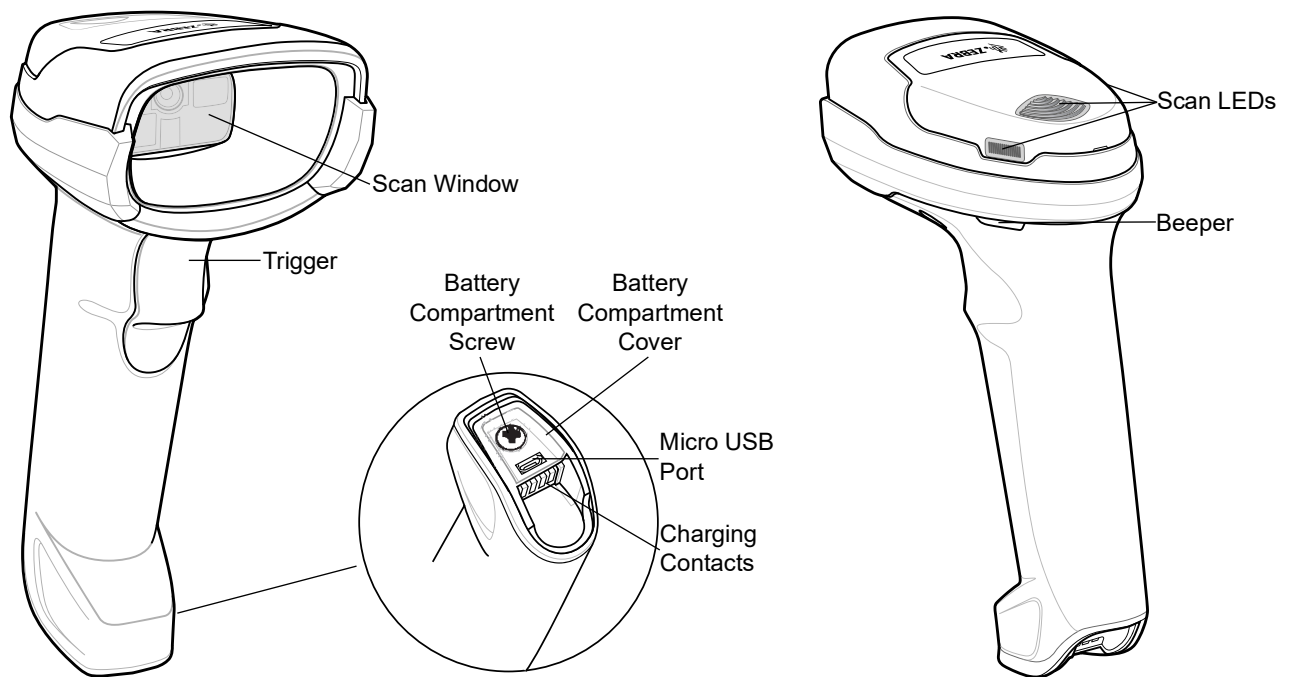


Figure 1-2 Digital Scanner Features

For detailed LED indicator information see [Beeper and LED Indications on page 3-1](#).

Cradle Features



NOTE For detailed information about connecting, using, and mounting the CR2278-PC cradle refer to the documentation included with the cradle (*CR2278-PC Presentation Cradle Quick Reference Guide*, p/n MN-002917xx).

The CR2278-PC cordless presentation cradle serves as a charger, radio communication interface, and host communication interface for the DS2278 cordless digital scanner.

The presentation cradle sits on a desktop and charges the DS2278 cordless digital scanner while allowing bar code scanning in presentation mode. This cradle also provides host communication by receiving digital scanner data via a Bluetooth radio, and sending that data to the host through an attached cable. The cable provides power to the cradle from the host or optional power supply if supported.

For more information about communication between the digital scanner, cradle, and host, see [Chapter 6, Radio Communications](#).

Presentation Cradle

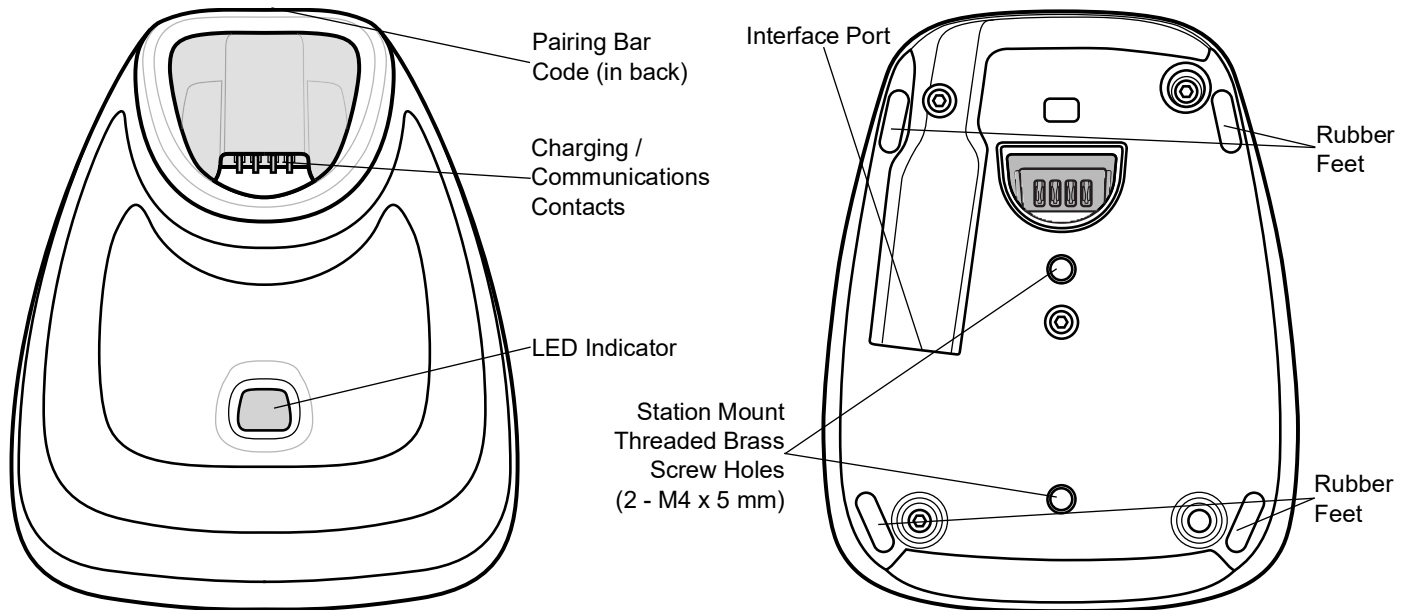


Figure 1-3 CR2278-PC (Presentation Cradle) -Top and Bottom Views

Connecting the Cradle

Via Host Connection (for Charging and Communication)

1. Connect the appropriate cable to the power supply and an AC power source, if necessary. This ensures detection of the host and prevents inadvertently back powering the cradle from improper detection of the host.
2. Insert the interface cable into the host port.
3. Insert the interface cable into the cradle's host port.
4. Pair the digital scanner to the cradle either by inserting it in the cradle (if Pair on Contacts is enabled) or by scanning the pairing bar code.
5. If necessary, scan the appropriate host bar code (for non-autodetected interfaces).

Via Wall Outlet Using a Power Adapter (for Charging Only)

1. Connect the appropriate cable to the power supply and an AC power source.
2. Insert the interface cable into the cradle's host port.
3. Pair the digital scanner directly to the host device (PC, Smartphone, or Tablet).

Changing the Host Interface

To connect to a different host, or to the same host through a different cable:

1. Unplug the interface cable from the host.
2. Unplug the power supply from the host cable, if required.
3. Connect the interface cable to the new host, or the new interface cable to the existing host.
4. Reconnect the power supply, if required.
5. If necessary, scan the appropriate host bar code (for non-autodetected interfaces).

Using a DC Power Supply

There is no separate power jack on the cradle. The cradle can operate from host supplied power, if available. If host power is limited or not available, an external DC power supply can be used with certain host interface cables that support a power jack. An external power supply is recommended if fastest charging is required. See host interface chapters for connections.



CAUTION Always disconnect the DC power supply BEFORE disconnecting the cable to the host end or the cradle may not recognize the new host.

Charging the DS2278 Battery

When using a new battery in the DS2278, the battery may require a charge to be enabled. See *Charging the DS2278 Battery on page 1-5*.



NOTES

1. It is recommended to fully charge the battery before first use.
2. Charge time depends on the host type and power source.
3. The CR2278-PC passes 5V to the digital scanner which charges the battery. The charging system on the DS2278 charges the battery with as much current as possible based on the power source of the CR2278-PC. This way the input power source is never overloaded and the charge time is made as short as possible. Charge times lengthen as scanning activity increases when a DS2278 operates in presentation mode on a CR2278-PC cradle. To optimize charge performance, ensure the digital scanner is oriented to minimize accidental scan activation.



CAUTION

The battery does not charge when the temperature is above 40°C (104°F). When scanning in presentation mode at temperatures above 40°C (104°F), the battery drains until the temperature drops. When the battery charge is consumed the digital scanner stops scanning until the temperature drops and the battery has sufficient time to charge.

To avoid temperature related faults, charge the battery and operate the DS2278 in presentation mode on a CR2278-PC only within the recommended 0°C to 40°C (32°F to 104°F) range, and ideally within 5°C to 35°C (41°F to 95°F).

Charging Using the Cradle

Insert the battery in the DS2278 and place the DS2278 in the CR2278-PC cradle (see *Inserting the Digital Scanner in the Cradle on page 1-9*). The digital scanner boots and the cradle LED begins flashing green to indicate the start of normal charging.

See [Beeper and LED Indications on page 3-1](#) for detailed LED indicator information.

Charging Using the Micro USB Cable

The Micro USB cable is a charge only cable.

✓ **NOTE** For best performance, we recommend using Zebra accessories.

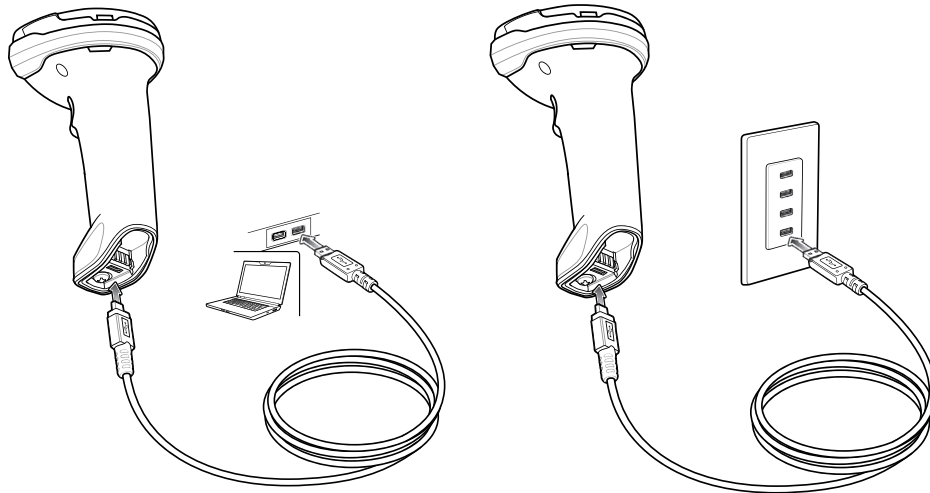


Figure 1-4 *Micro USB Connections*

Connect the Micro USB connector to the DS2278. Connect the standard USB connector to a PC or a USB wall outlet. The digital scanner enumerates as a CDC device when the USB connector is plugged into a PC host. When charging begins, the digital scanner LED begins flashing green to indicate the start of normal charging.

Converting a CDC Device to HID

The digital scanner enumerates over USB as a CDC device, by default. If your host does not have a CDC driver, a delay occurs between connecting the digital scanner to the host and the start of charging. To eliminate the delay scan **HID Device Conversion** below to enumerate as an HID device. This allows the scanner to begin charging sooner.

To switch the digital scanner back to a CDC device, scan **Scanner as CDC Device** below.



HID Device Conversion

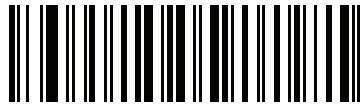


***Scanner as CDC Device**

For instructions on downloading CDC drivers see [page 8-6](#).

Shutting Off the Digital Scanner Battery

Scan the **Battery Off** bar code below to shut off the battery for long term storage or shipping. Pull the digital scanner trigger to turn on the battery.



Battery Off

- ✓ **NOTE** Always scan the **Battery Off** bar code in hand-held mode. Also see [Battery Preservation Mode on page 5-15](#) for additional battery features.

Inserting the Battery

- ✓ **NOTES**
1. It is recommended to fully charge the battery before first use.
 2. The battery is initially shipped inside the digital scanner. Inserting a battery would only be applicable for a replacement battery.

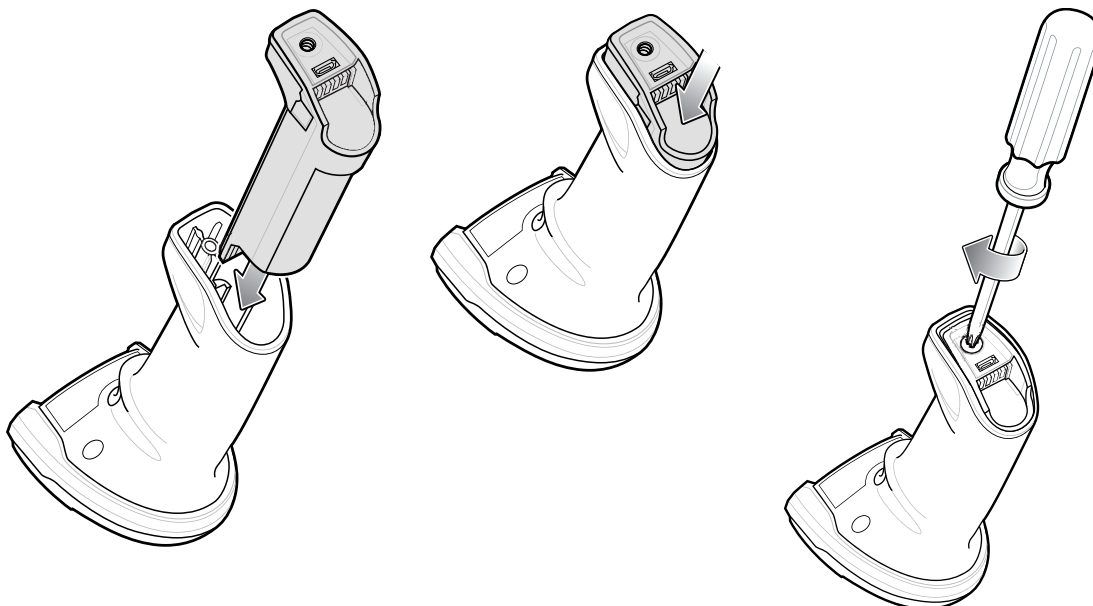


Figure 1-5 *Inserting the Battery*

To insert the battery into the digital scanner:

1. Insert the battery into the battery compartment and push down.
2. Turn the screw on the battery compartment clockwise to tighten. Do not over torque the screw.